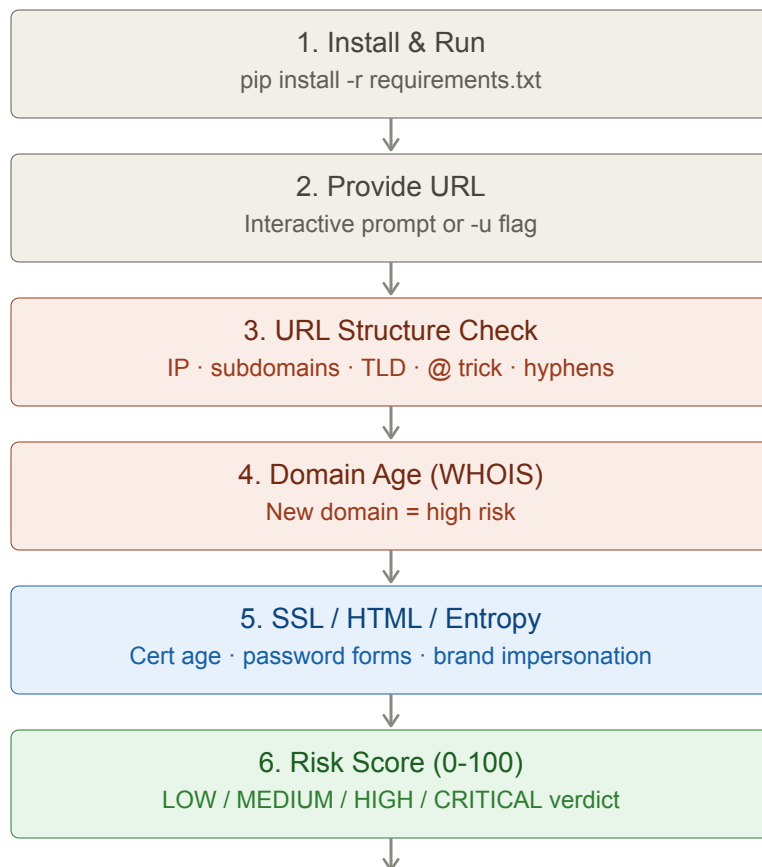


Phishing Detector — cheat sheet

Analyzes any URL and returns a risk score (0–100) covering domain structure, age, SSL, HTML content, and brand impersonation.



Step 1 — Install & Run

```
pip install -r requirements.txt

# Interactive mode
python3 phishing_detector.py

# Direct URL
python3 phishing_detector.py -u https://suspicious-site.com

# With VirusTotal API key (optional)
python3 phishing_detector.py -u https://site.com --vt-key YOUR_API_KEY
```

Step 2 — Six Checks Explained

Check	What it looks for	Red flags
URL Structure	Domain anatomy	Raw IP, excessive subdomains, @ trick, suspicious TLD
Domain Age	WHOIS registration date	Domain registered under 30 days ago
SSL Cert	Certificate details	Free CA, no HTTPS, newly issued cert, domain mismatch
HTML Content	Page source analysis	Password forms to external domains, phishing keywords
Domain Entropy	Randomness + typosquat	High entropy (DGA), paypa1, g00gle substitutions
VirusTotal	Vendor reputation	Any vendor flagged the URL (requires API key)

Step 3 — Reading the Score

URL: `https://paypal.secure-verify.tk/login`

Check Results:

URL Structure	: HIGH	(25/25)	raw IP, suspicious TLD
Domain Age	: HIGH	(20/20)	registered 3 days ago
SSL Certificate	: MEDIUM	(10/15)	free CA, 2 days old
HTML Content	: HIGH	(20/20)	password form + phishing keywords
Domain Entropy	: MEDIUM	(10/10)	typosquat: paypa1 ~ paypal
VirusTotal	: -		(no API key)

FINAL RISK SCORE : 85/100

VERDICT : CRITICAL — Very likely phishing

Score	Verdict
0-30	LOW — likely safe
31-50	MEDIUM — investigate further
51-70	HIGH — likely malicious
71-100	CRITICAL — very likely phishing

Step 4 — VirusTotal Integration

```
# Get a free API key at virustotal.com
python3 phishing_detector.py -u https://site.com --vt-key abc123xyz

# Or set as env var
export VT_API_KEY=abc123xyz
python3 phishing_detector.py -u https://site.com
```

Key idea: a newly registered domain (under 30 days) with HTTPS and no VirusTotal flags can still be a phishing page — SSL is free and easy. Domain age + HTML content analysis catch these early-stage pages.